

Problem 9

From a table of random numbers, one number is randomly selected. If the selected number is divisible by 5, then this is event A ; if that number ends with zero, then this is event B . What does event $A \setminus B$ mean?

Solution

We need to interpret the set difference $A \setminus B$ in the context of divisibility.

Step 1: Define the Events

Event A

Event A = "the selected number is divisible by 5"

A number is divisible by 5 if and only if it ends in 0 or 5.

Examples: 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, ...

Event B

Event B = "the selected number ends with zero"

A number ends with zero if its last digit is 0.

Examples: 0, 10, 20, 30, 40, 50, 60, 70, 80, 90, 100, ...

Step 2: Relationship Between Events

Observation 1: $B \subset A$

If a number ends with zero, then it is divisible by 5.

This is because any number ending in 0 can be written as $10k$ for some integer k , and $10k = 5 \cdot (2k)$ is divisible by 5.

Therefore: $B \subset A$

Observation 2: A Contains More Than B

Event A includes:

- Numbers ending in 0 (which are in B)
- Numbers ending in 5 (which are NOT in B)

Examples of numbers in A but not in B : 5, 15, 25, 35, 45, 55, 65, 75, 85, 95, ...

Step 3: Interpret $A \setminus B$ (Set Difference)

Definition

The set difference $A \setminus B$ (also written as $A - B$ or $A \div B$) is defined as:

$$A \setminus B = \{x : x \in A \text{ and } x \notin B\}$$

This represents elements that are in A but not in B .

Application to Our Events

$$A \setminus B = \{\text{numbers that are divisible by 5 but do not end with zero}\}$$

Step 4: Characterize $A \setminus B$

Which Numbers Are in $A \setminus B$?

A number is in $A \setminus B$ if:

- It is divisible by 5 (so it ends in 0 or 5), AND
- It does NOT end with zero

Therefore, a number is in $A \setminus B$ if and only if:

The number ends with 5

Step 5: Final Answer

$A \setminus B =$ "The selected number ends with the digit 5"

Equivalently:

$A \setminus B =$ "The number is divisible by 5 but not divisible by 10"

Step 6: Examples

Numbers in Event A

$A = \{5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, 100, \dots\}$

All numbers ending in 0 or 5.

Numbers in Event B

$B = \{0, 10, 20, 30, 40, 50, 60, 70, 80, 90, 100, 110, \dots\}$

All numbers ending in 0 (multiples of 10).

Numbers in Event $A \setminus B$

$A \setminus B = \{5, 15, 25, 35, 45, 55, 65, 75, 85, 95, 105, 115, \dots\}$

All numbers ending in 5 (multiples of 5 that are not multiples of 10).

Step 7: Set Notation Verification

We can verify using set operations:

$$A = B \cup (A \setminus B)$$

Indeed:

- B = numbers ending in 0
- $A \setminus B$ = numbers ending in 5
- $B \cup (A \setminus B)$ = numbers ending in 0 or 5 = numbers divisible by 5 = A

Also:

$$A \setminus B = A \cap \bar{B}$$

where $\bar{B}$ = numbers that do NOT end with zero.

Step 8: Alternative Characterizations

Event $A \setminus B$ can be described in several equivalent ways:

1. The number ends with digit 5
2. The number is an odd multiple of 5
3. The number is divisible by 5 but not by 10
4. The number is of the form $5(2k + 1)$ for some non-negative integer k
5. The number is congruent to 5 modulo 10

Step 9: Probability Consideration

If we're selecting from single-digit numbers (0-9):

- $A = \{0, 5\}$, so $P(A) = \frac{2}{10} = 0.2$
- $B = \{0\}$, so $P(B) = \frac{1}{10} = 0.1$
- $A \setminus B = \{5\}$, so $P(A \setminus B) = \frac{1}{10} = 0.1$

In general, for uniformly distributed digits:

$$P(A \setminus B) = P(A) - P(B) = 0.2 - 0.1 = 0.1$$

Final Answer

$A \setminus B = \text{"The selected number ends with the digit 5"}$
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Equivalent statements:

- The number is divisible by 5 but not by 10
- The number is an odd multiple of 5
- The number has the form $10k + 5$ for some non-negative integer k

Examples: 5, 15, 25, 35, 45, 55, 65, 75, 85, 95, 105, ...

Key insight: Since $B \subset A$ (every number ending in 0 is divisible by 5), the set difference $A \setminus B$ consists of the numbers in A that are not in B , which are precisely the numbers ending in 5.